Chapter 3

Introduction to Sobolev Spaces

Remark 3.1 *Contents.* Sobolev spaces are the basis of the theory of weak or variational forms of partial differential equations. A very popular approach for discretizing partial differential equations, the finite element method, is based on variational forms. In this chapter, a short introduction into Sobolev spaces will be given. Recommended literature are the books Adams (1975); Adams and Fournier (2003) and Evans (2010). □

3.1 Elementary Inequalities

Lemma 3.2 Inequality for strictly monotonically increasing function. *Let* $f : \mathbb{R}_+ \cup \{0\} \to \mathbb{R}$ *be a continuous and strictly monotonically increasing function with* $f(0) = 0$ *and* $f(x) \to \infty$ *for* $x \to \infty$*. Then, for all* $a, b \in \mathbb{R}_+ \cup \{0\}$ *it is*

$$ab \leq \int_0^a f(x)\ dx + \int_0^b f^{-1}(y)\ dy,$$

where $f^{-1}(y)$ *is the inverse of* $f(x)$*.*

Proof: Since $f(x)$ is strictly monotonically increasing, the inverse function exists. The proof is based on a geometric argument, see Figure 3.1.

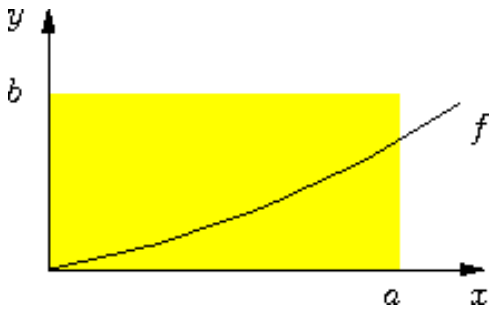


Figure 3.1: Sketch to the proof of Lemma 3.2.

Consider the interval $(0, a)$ on the x-axis and the interval $(0, b)$ on the y-axis. Then, the area of the corresponding rectangle is given by ab, $\int_0^a f(x)\ dx$ is the area below the curve, and $\int_0^b f^{-1}(y)\ dy$ is the area between the positive y-axis and the curve. From Figure 3.1, the inequality follows immediately. The equal sign holds only iff $f(a) = b$. ■

Remark 3.3 *Young's inequality.* Young's inequality

$$ab \leq \frac{\varepsilon}{2}a^2 + \frac{1}{2\varepsilon}b^2 \quad \forall\ a, b, \varepsilon \in \mathbb{R}_+ \tag{3.1}$$

follows from Lemma 3.2 with $f(x) = \varepsilon x$, $f^{-1}(y) = \varepsilon^{-1}y$. It is also possible to derive this inequality from the binomial theorem. For proving the generalized Young inequality

$$ab \leq \frac{\varepsilon^p}{p}a^p + \frac{1}{q\varepsilon^q}b^q, \quad \forall\ a, b, \varepsilon \in \mathbb{R}_+ \tag{3.2}$$

with $p^{-1} + q^{-1} = 1, p, q \in (1, \infty)$, one chooses $f(x) = x^{p-1}$, $f^{-1}(y) = y^{1/(p-1)}$ and applies Lemma 3.2 with intervals where the upper bounds are given by εa and $\varepsilon^{-1}b$. □

Remark 3.4 *Cauchy–Schwarz inequality.* The Cauchy[1]–Schwarz[2] inequality (for vectors, for sums)

$$|(\mathbf{x}, \mathbf{y})| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \ \forall\ \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \tag{3.3}$$

where $(\cdot, \cdot)$ is the Euclidean product and $\|\cdot\|_2$ the Euclidean norm, is well known. One can prove this inequality with the help of Young's inequality.

First, it is clear that the Cauchy–Schwarz inequality is correct if one of the vectors is the zero vector. Now, let $\mathbf{x}, \mathbf{y}$ with $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 = 1$. One obtains with the triangle inequality and Young's inequality (3.1)

$$|(\mathbf{x}, \mathbf{y})| = \left|\sum_{i=1}^n x_i y_i\right| \leq \sum_{i=1}^n |x_i|\,|y_i| \leq \frac{1}{2}\sum_{i=1}^n |x_i|^2 + \frac{1}{2}\sum_{i=1}^n |y_i|^2 = 1.$$

Hence, the Cauchy–Schwarz inequality is correct for $\mathbf{x}, \mathbf{y}$. Last, one considers arbitrary vectors $\tilde{\mathbf{x}} \neq \mathbf{0}, \tilde{\mathbf{y}} \neq \mathbf{0}$. Now, one can utilize the homogeneity of the Cauchy–Schwarz inequality. From the validity of the Cauchy–Schwarz inequality for $\mathbf{x}$ and $\mathbf{y}$, one obtains by a scaling argument

$$\big|(\underbrace{\|\tilde{\mathbf{x}}\|_2^{-1}\tilde{\mathbf{x}}}_{\mathbf{x}}, \underbrace{\|\tilde{\mathbf{y}}\|_2^{-1}\tilde{\mathbf{y}}}_{\mathbf{y}})\big| \leq 1$$

Both vectors $\mathbf{x}, \mathbf{y}$ have the Euclidean norm 1, hence

$$\frac{1}{\|\tilde{\mathbf{x}}\|_2 \|\tilde{\mathbf{y}}\|_2}\,|(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})| \leq 1 \quad \Longleftrightarrow \quad |(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})| \leq \|\tilde{\mathbf{x}}\|_2 \|\tilde{\mathbf{y}}\|_2\,.$$

The generalized Cauchy–Schwarz inequality or Hölder inequality

$$|(\mathbf{x}, \mathbf{y})| \leq \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} \left(\sum_{i=1}^n |y_i|^q\right)^{1/q}$$

with $p^{-1} + q^{-1} = 1, p, q \in (1, \infty)$, can be proved in the same way with the help of the generalized Young inequality. □

Definition 3.5 *Lebesgue spaces.* The space of functions which are Lebesgue integrable on Ω to the power of $p \in [1, \infty)$ is denoted by

$$L^p(\Omega) = \left\{ f \ :\ \int_\Omega |f|^p(\mathbf{x})\ d\mathbf{x} < \infty \right\},$$

[1] Augustin Louis Cauchy (1789 – 1857)
[2] Hermann Amandus Schwarz (1843 – 1921)

which is equipped with the norm

$$\|f\|_{L^p(\mathbf{x})} = \left(\int_\Omega |f|^p(\mathbf{x}) \; d\mathbf{x} \right)^{1/p}.$$

For $p = \infty$, this space is

$$L^\infty(\Omega) = \{f \; : \; |f(\mathbf{x})| < \infty \text{ almost everywhere in } \Omega\}$$

with the norm

$$\|f\|_{L^\infty(\Omega)} = \text{ess sup}_{\mathbf{x}\in\Omega} |f(\mathbf{x})|.$$

□

Lemma 3.6 Hölder's inequality. *Let $p^{-1} + q^{-1} = 1, p, q \in [1, \infty]$. If $u \in L^p(\Omega)$ and $v \in L^q(\Omega)$, then it is $uv \in L^1(\Omega)$ and it holds that*

$$\|uv\|_{L^1(\Omega)} \leq \|u\|_{L^p(\Omega)} \|v\|_{L^q(\Omega)}. \tag{3.4}$$

If $p = q = 2$, then this inequality is also known as Cauchy–Schwarz inequality

$$\|uv\|_{L^1(\Omega)} \leq \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)}. \tag{3.5}$$

Proof: $p, q \in (1, \infty)$. First, one has to show that $|uv(\mathbf{x})|$ can be estimated from above by an integrable function. Setting in the generalized Young inequality (3.2) $\varepsilon = 1$, $a = |u(\mathbf{x})|$, and $b = |v(\mathbf{x})|$ gives

$$|u(\mathbf{x})v(\mathbf{x})| \leq \frac{1}{p} |u(\mathbf{x})|^p + \frac{1}{q} |v(\mathbf{x})|^q.$$

Since the right hand side of this inequality is integrable, by assumption, it follows that $uv \in L^1(\Omega)$. In addition, Hölder's inequality is proved for the case $\|u\|_{L^p(\Omega)} = \|v\|_{L^q(\Omega)} = 1$ using this inequality

$$\int_\Omega |u(\mathbf{x})v(\mathbf{x})| \; d\mathbf{x} \leq \frac{1}{p} \int_\Omega |u(\mathbf{x})|^p \; d\mathbf{x} + \frac{1}{q} \int_\Omega |v(\mathbf{x})|^q \; d\mathbf{x} = 1.$$

The general inequality follows, for the case that both functions do not vanish almost everywhere, with the same homogeneity argument as used for proving the Cauchy–Schwarz inequality of sums. In the case that one of the functions vanishes almost everywhere, (3.4) is trivially satisfied.

$p = 1, q = \infty$. It is

$$\int_\Omega |u(\mathbf{x})v(\mathbf{x})| \; d\mathbf{x} \leq \int_\Omega |u(\mathbf{x})| \, \text{ess sup}_{\mathbf{x}\in\Omega} |v(\mathbf{x})| \; d\mathbf{x} = \|u\|_{L^1(\Omega)} \|v\|_{L^\infty(\Omega)}.$$

■

3.2 Weak Derivative and Distributions

Remark 3.7 *Contents.* This section introduces a generalization of the derivative which is needed for the definition of weak or variational problems. For an introduction to the topic of this section, see, e.g., Haroske and Triebel (2008)

Let $\Omega \subset \mathbb{R}^d$ be a domain with boundary $\Gamma = \partial\Omega$, $d \in \mathbb{N}$, $\Omega \neq \emptyset$. A domain is always an open set. □

Definition 3.8 The space $C_0^\infty(\Omega)$. The space of infinitely often differentiable real functions with compact (closed and bounded) support in Ω is denoted by $C_0^\infty(\Omega)$

$$C_0^\infty(\Omega) = \{v \; : \; v \in C^\infty(\Omega), \; \text{supp}(v) \subset \Omega\},$$

where

$$\text{supp}(v) = \overline{\{\mathbf{x} \in \Omega \; : \; v(\mathbf{x}) \neq 0\}}.$$

□

Definition 3.9 Convergence in $C_0^\infty(\Omega)$. The sequence of functions $\{\phi_n(\mathbf{x})\}_{n=1}^\infty$, $\phi_n \in C_0^\infty(\Omega)$ for all n, is said to convergence to the zero functions if and only if
a) $\exists K \subset \Omega, K$ compact (closed and bounded) with $\text{supp}(\phi_n) \subset K$ for all n,
b) $D^{\boldsymbol{\alpha}}\phi_n(\mathbf{x}) \to 0$ for $n \to \infty$ on K for all multi-indices $\alpha = (\alpha_1, \ldots, \alpha_d)$, $|\alpha| = \alpha_1 + \ldots + \alpha_d$.
It is

$$\lim_{n\to\infty} \phi_n(\mathbf{x}) = \phi(\mathbf{x}) \quad \Longleftrightarrow \quad \lim_{n\to\infty} (\phi_n(\mathbf{x}) - \phi(\mathbf{x})) = 0.$$

□

Definition 3.10 Weak derivative. Let $f, F \in L^1_{\text{loc}}(\Omega)$. ($L^1_{\text{loc}}(\Omega)$: for each compact subset $\Omega' \subset \Omega$ it holds

$$\int_{\Omega'} |u(\mathbf{x})| \; d\mathbf{x} < \infty \; \forall \; u \in L^1_{\text{loc}}(\Omega).)$$

If for all functions $g \in C_0^\infty(\Omega)$ it holds that

$$\int_\Omega F(\mathbf{x}) g(\mathbf{x}) \; d\mathbf{x} = (-1)^{|\boldsymbol{\alpha}|} \int_\Omega f(\mathbf{x}) D^{\boldsymbol{\alpha}} g(\mathbf{x}) \; d\mathbf{x},$$

then $F(\mathbf{x})$ is called weak derivative of $f(\mathbf{x})$ with respect to the multi-index $\boldsymbol{\alpha}$. □

Remark 3.11 *On the weak derivative.*
- One uses the same notations for the derivative as in the classical case : $F(\mathbf{x}) = D^{\boldsymbol{\alpha}} f(\mathbf{x})$.
- If $f(\mathbf{x})$ is classically differentiable on Ω, then the classical derivative is also the weak derivative.
- The assumptions on $f(\mathbf{x})$ and $F(\mathbf{x})$ are such that the integrals in the definition of the weak derivative are well defined. In particular, since the test functions vanish in a neighborhood of the boundary, the behavior of $f(\mathbf{x})$ and $F(\mathbf{x})$ if $\mathbf{x}$ approaches the boundary is not of importance.
- The main aspect of the weak derivative is due to the fact that the (Lebesgue) integral is not influenced from the values of the functions on a set of (Lebesgue) measure zero. Hence, the weak derivative is uniquely defined only up to a set of measure zero. It follows that $f(\mathbf{x})$ might be not classically differentiable on a set of measure zero, e.g., in a point, but it can still be weakly differentiable.
- The weak derivative is uniquely determined, in the sense described above.

□

Example 3.12 *Weak derivative.* The weak derivative of the function $f(x) = |x|$ is

$$f'(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0 \end{cases}$$

In $x = 0$, one can use also any other real number. The proof of this statement follows directly from the definition and it is left as an exercise. □

Definition 3.13 Distribution. A continuous linear functional defined on $C_0^\infty(\Omega)$ is called distribution. The set of all distributions is denoted by $(C_0^\infty(\Omega))'$.

Let $u \in C_0^\infty(\Omega)$ and $\psi \in (C_0^\infty(\Omega))'$, then the following notation is used for the application of the distribution to the function

$$\psi(u(\mathbf{x})) = \langle \psi, u \rangle \in \mathbb{R}.$$

□

Remark 3.14 *On distributions.* Distributions are a generalization of functions. They assign each function from $C_0^\infty(\Omega)$ a real number. □

Example 3.15 *Regular distribution.* Let $u(\mathbf{x}) \in L^1_{\text{loc}}(\Omega)$. Then, a distribution is defined by

$$\int_\Omega u(\mathbf{x})\phi(\mathbf{x})\ d\mathbf{x} = \langle \psi, \phi \rangle,\ \forall \phi \in C_0^\infty(\Omega).$$

This distribution will be identified with $u(\mathbf{x}) \in L^1_{\text{loc}}(\Omega)$.

Distributions with such an integral representation are called regular, otherwise they are called singular. □

Example 3.16 *Dirac distribution.* Let $\boldsymbol{\xi} \in \Omega$ fixed, then

$$\langle \delta_{\boldsymbol{\xi}}, \phi \rangle = \phi(\boldsymbol{\xi})\ \forall\ \phi \in C_0^\infty(\Omega)$$

defines a singular distribution, the so-called Dirac distribution or δ-distribution. It is denoted by $\delta_{\boldsymbol{\xi}} = \delta(\mathbf{x} - \boldsymbol{\xi})$. □

Definition 3.17 *Derivatives of distributions.* Let $\phi \in (C_0^\infty(\Omega))'$ be a distribution. The distribution $\psi \in (C_0^\infty(\Omega))'$ is called derivative in the sense of distributions or distributional derivative of ϕ if

$$\langle \psi, u \rangle = (-1)^{|\boldsymbol{\alpha}|} \langle \phi, D^{\boldsymbol{\alpha}} u \rangle\ \forall u \in C_0^\infty(\Omega),$$

$\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d)$, $\alpha_j \geq 0, j = 1, \ldots, d$, $|\boldsymbol{\alpha}| = \alpha_1 + \ldots + \alpha_d$. □

Remark 3.18 *On derivatives of distributions.* Each distribution has derivatives in the sense of distributions of arbitrary order.

If the derivative in the sense of distributions $D^\alpha u(\mathbf{x})$ of $u(\mathbf{x}) \in L^1_{\text{loc}}(\Omega)$ is a regular distribution, then also the weak derivative of $u(\mathbf{x})$ exists and both derivatives are identified. □

3.3 Lebesgue Spaces and Sobolev Spaces

Remark 3.19 *On the spaces $L^p(\Omega)$.* These spaces were introduced in Definition 3.5.

- The elements of $L^p(\Omega)$ are, strictly speaking, equivalence classes of functions which are different only on a set of Lebesgue measure zero.
- The spaces $L^p(\Omega)$ are Banach spaces (complete normed spaces). A space X is complete, if each so-called Cauchy sequence $\{u_n\}_{n=0}^\infty \in X$, i.e., for all $\varepsilon > 0$ there is an index $n_0(\varepsilon)$ such that for all $i, j > n_0(\varepsilon)$

  $$\|u_i - u_j\|_X < \varepsilon.$$

  converges and the limit is an element of X.
- The space $L^2(\Omega)$ becomes a Hilbert spaces with the inner product

  $$(f, g) = \int_\Omega f(\mathbf{x})g(\mathbf{x})\ d\mathbf{x}, \quad \|f\|_{L^2} = (f, f)^{1/2}, \quad f, g \in L^2(\Omega).$$

- The dual space of a space X is the space of all bounded linear functionals defined on X. Let Ω be a domain with sufficiently smooth boundary Γ. of the Lebesgue spaces $L^p(\Omega)$, $p \in [1, \infty]$, then

  $$\begin{aligned} (L^p(\Omega))' &= L^q(\Omega) \ \text{ with } \ p, q \in (1, \infty),\ \frac{1}{p} + \frac{1}{q} = 1, \\ \left(L^1(\Omega)\right)' &= L^\infty(\Omega), \\ (L^\infty(\Omega))' &\neq L^1(\Omega). \end{aligned}$$

The spaces $L^1(\Omega)$, $L^\infty(\Omega)$ are not reflexive, i.e., the dual space of the dual space is not the original space again.

□

Definition 3.20 Sobolev[3] spaces. Let $k \in \mathbb{N} \cup \{0\}$ and $p \in [1, \infty]$, then the Sobolev space $W^{k,p}(\Omega)$ is defined by

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) \ : \ D^{\boldsymbol{\alpha}} u \in L^p(\Omega) \ \forall \ \boldsymbol{\alpha} \text{ with } |\boldsymbol{\alpha}| \leq k\}.$$

This space is equipped with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \sum_{|\boldsymbol{\alpha}| \leq k} \|D^{\boldsymbol{\alpha}} u\|_{L^p(\Omega)} \,. \tag{3.6}$$

□

Remark 3.21 *On the spaces $W^{k,p}(\Omega)$.*

- Definition 3.20 has the following meaning. From $u \in L^p(\Omega)$, $p \in [1, \infty)$, it follows in particular that $u \in L^1_{\text{loc}}(\Omega)$, such that $u(\mathbf{x})$ defines (represents) a distribution. Then, all derivatives $D^{\boldsymbol{\alpha}} u$ exist in the sense of distributions. The statement $D^{\boldsymbol{\alpha}} u \in L^p(\Omega)$ means that the distribution $D^{\boldsymbol{\alpha}} u \in (C_0^\infty(\Omega))'$ can be represented by a function from $L^p(\Omega)$.
- One can add elements from $W^{k,p}(\Omega)$ and one can multiply them with real numbers. The result is again a function from $W^{k,p}(\Omega)$. With this property, the space $W^{k,p}(\Omega)$ becomes a vector space (linear space). It is straightforward to check that (3.6) is a norm. (*exercise*)
- It is $D^{\boldsymbol{\alpha}} u(\mathbf{x}) = u(\mathbf{x})$ for $\boldsymbol{\alpha} = (0, \ldots, 0)$ and $W^{0,p}(\Omega) = L^p(\Omega)$.
- The spaces $W^{k,p}(\Omega)$ are Banach spaces.
- Sobolev spaces have for $p \in [1, \infty)$ a countable basis $\{\varphi_n(\mathbf{x})\}_{n=1}^\infty$ (Schauder basis), i.e., each element $u(\mathbf{x})$ can be written in the form

$$u(\mathbf{x}) = \sum_{n=1}^{\infty} u_n \varphi_n(\mathbf{x}), \quad u_n \in \mathbb{R} \ n = 1, \ldots, \infty.$$

- Sobolev spaces are uniformly convex for $p \in (1, \infty)$, i.e., for each $\varepsilon \in (0, 2]$ (note that the largest distance in the ball is equal to 2) there is a $\delta(\varepsilon) > 0$ such that for all $u, v \in W^{k,p}(\Omega)$ with $\|u\|_{W^{k,p}(\Omega)} = \|v\|_{W^{k,p}(\Omega)} = 1$, and $\|u - v\|_{W^{k,p}(\Omega)} > \varepsilon$ it holds that $\left\|\frac{u+v}{2}\right\|_{W^{k,p}(\Omega)} \leq 1 - \delta(\varepsilon)$, see Figure 3.2 for an illustration.
- Sobolev spaces are reflexive for $p \in (1, \infty)$.
- On can show that $C^\infty(\Omega)$ is dense in $W^{k,p}(\Omega)$, e.g., see (Alt, 1999, Satz 1.21, Satz 2.10) or (Adams, 1975, Lemma 3.15). With this property, one can characterize the Sobolev spaces $W^{k,p}(\Omega)$ as completion of the functions from $C^\infty(\Omega)$ with respect to the norm (3.6). For domains with smooth boundary, one can even show that $C^\infty(\overline{\Omega})$ is dense in $W^{k,p}(\Omega)$.
- The Sobolev space $H^k(\Omega) = W^{k,2}(\Omega)$ is a Hilbert space with the inner product

$$(u, v)_{H^k(\Omega)} = \sum_{|\boldsymbol{\alpha}| \leq k} \int_\Omega D^{\boldsymbol{\alpha}} u(\mathbf{x}) D^{\boldsymbol{\alpha}} v(\mathbf{x}) \, d\mathbf{x}$$

  and the norm $\|u\|_{H^k(\Omega)} = (u, u)^{1/2}$.

□

[3] Sergei Lvovich Sobolev (1908 – 1989)

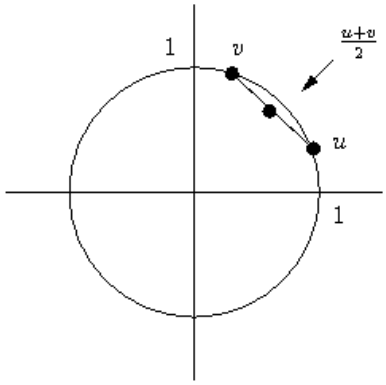


Figure 3.2: Illustration of the uniform convexity of Sobolev spaces.

Definition 3.22 The space $W_0^{k,p}(\Omega)$. The Sobolev space $W_0^{k,p}(\Omega)$ is defined as the completion of $C_0^\infty(\Omega)$ in the norm of $W^{k,p}(\Omega)$

$$W_0^{k,p}(\Omega) = \overline{C_0^\infty(\Omega)}^{\|\cdot\|_{W^{k,p}(\Omega)}}.$$

□

3.4 The Trace of a Function from a Sobolev Space

Remark 3.23 *Motivation.* This class considers boundary value problems for partial differential equations. In the theory of weak or variational solutions, the solution of the partial differential equation is searched in an appropriate Sobolev space. Then, for the boundary value problem, this solution has to satisfy the boundary condition. However, since the boundary of a domain is a manifold of dimension $(d-1)$, and consequently it has Lebesgue measure zero, one has to clarify how a function from a Sobolev space is defined on this manifold. This definition will be presented in this section. □

Definition 3.24 Boundary of class $C^{k,\alpha}$. A bounded domain $\Omega \subset \mathbb{R}^d$ and its boundary Γ are of class $C^{k,\alpha}$, $0 \le \alpha \le 1$ if for all $\mathbf{x}_0 \in \Gamma$ there is a ball $B(\mathbf{x}_0, r)$ with $r > 0$ and a bijective map $\psi : B(\mathbf{x}_0, r) \to D \subset \mathbb{R}^d$ such that
1) $\psi\left(B(\mathbf{x}_0, r) \cap \Omega\right) \subset \mathbb{R}^d_+$,
2) $\psi\left(B(\mathbf{x}_0, r) \cap \Gamma\right) \subset \partial\mathbb{R}^d_+$,
3) $\psi \in C^{k,\alpha}(B(\mathbf{x}_0, r)), \psi^{-1} \in C^{k,\alpha}(D)$, are Hölder continuous.
That means, Γ is locally the graph of a function with $d-1$ arguments. (A function $u(\mathbf{x})$ is Hölder continuous if

$$\|u\|_{C^{k,\alpha}(\Omega)} = \sum_{|\boldsymbol{\alpha}| \le k} \|D^{\boldsymbol{\alpha}} u\|_{C(\overline{\Omega})} + \sum_{|\boldsymbol{\alpha}| = k} [D^{\boldsymbol{\alpha}} u]_{C^{0,\alpha}(\overline{\Omega})}$$

with

$$[D^{\boldsymbol{\alpha}} u]_{C^{0,\alpha}(\overline{\Omega})} = \sup_{\mathbf{x},\mathbf{y} \in \Omega} \left\{ \frac{|u(\mathbf{x}) - u(\mathbf{y})|}{|\mathbf{x} - \mathbf{y}|^\alpha} \right\}$$

is finite.) □

Remark 3.25 *Lipschitz boundary.* It will be generally assumed that the boundary of Ω is of class $C^{0,1}$. That means, the map is Lipschitz[4] continuous. Such a boundary

[4]Rudolf Otto Sigismund Lipschitz (1832 – 1903)

is simply called Lipschitz boundary and the domain is called Lipschitz domain. An important feature of a Lipschitz boundary is that the outer normal vector is defined almost everywhere at the boundary and it is almost everywhere continuous. □

Example 3.26 *On Lipschitz domains.*

- Domains with Lipschitz boundary are, for example, balls or polygonal domains in two dimensions where the domain is always on one side of the boundary.
- A domain which is not a Lipschitz domain is a circle with a slit

$$\Omega = \{(x, y) \ : \ x^2 + y^2 < 1\} \setminus \{(x, y) \ : \ x \geq 0, y = 0\}.$$

  At the slit, the domain is on both sides of the boundary.
- In three dimension, a polyhedral domain is not not necessarily a Lipschitz domain. For instance, if the domain is build of two bricks which are laying on each other like in Figure 3.3, then the boundary is not Lipschitz continuous where the edge of one brick meets the edge of the other brick.

□

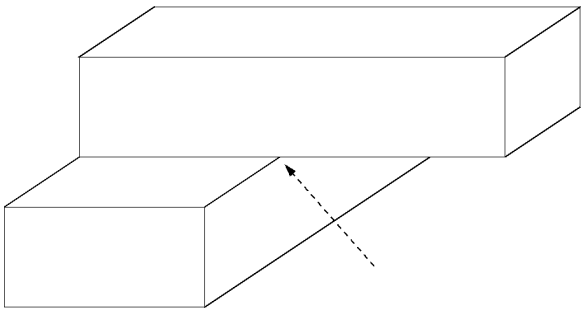


Figure 3.3: Polyhedral domain in three dimensions which is not Lipschitz continuous (at the corner where the arrow points to).

Theorem 3.27 Trace theorem. *Let $\Omega \subset \mathbb{R}^d$, $d \geq 2$, with a Lipschitz boundary. Then, there is exactly one linear and continuous operator $\gamma : W^{1,p}(\Omega) \to L^p(\Gamma)$, $p \in [1, \infty)$, which gives for functions $u \in C(\overline{\Omega}) \cap W^{1,p}(\Omega)$ the classical boundary values*

$$\gamma u(\mathbf{x}) = u(\mathbf{x}), \ \mathbf{x} \in \Gamma, \ \forall \ u \in C(\overline{\Omega}) \cap W^{1,p}(\Omega),$$

i.e., $\gamma u(\mathbf{x}) = u(\mathbf{x})|_{\mathbf{x} \in \Gamma}$.

Proof: The proof can be found in the literature, e.g., in Adams (1975); Adams and Fournier (2003). ■

Remark 3.28 *On the trace.* The operator γ is called trace or trace operator.

- Since a linear and continuous operator is bounded, there is a constant $C > 0$ with

$$\|\gamma u\|_{L^p(\Gamma)} \leq C \, \|u\|_{W^{1,p}(\Omega)} \ \forall \ u \in W^{1,p}(\Omega)$$

  or

$$\|\gamma\|_{\mathcal{L}(W^{1,p}(\Omega), L^p(\Gamma))} \leq C.$$

- By definition of the trace, one gets for $u \in C(\overline{\Omega})$ the classical boundary values. By the density of $C^\infty(\overline{\Omega})$ in $W^{1,p}(\Omega)$ for domains with smooth boundary, it follows that $C(\overline{\Omega})$ is also dense in $W^{1,p}(\Omega)$ such that for all $u \in W^{1,p}(\Omega)$ there is a sequence $\{u_n\}_{n=1}^\infty \in C(\overline{\Omega})$ with $u_n \to u$ in $W^{1,p}(\Omega)$. Then, the trace of u is defined to be $\gamma u = \lim_{k \to \infty}(\gamma u_k)$.

- It is

$$\begin{aligned} \gamma u(\mathbf{x}) &= 0 \quad \forall\, u \in W_0^{1,p}(\Omega), \\ \gamma D^{\boldsymbol{\alpha}} u(\mathbf{x}) &= 0 \quad \forall\, u \in W_0^{k,p}(\Omega), |\boldsymbol{\alpha}| \le k-1. \end{aligned} \tag{3.7}$$

□

3.5 Sobolev Spaces with Non-Integer and Negative Exponents

Remark 3.29 *Motivation.* Sobolev spaces with non-integer and negative exponents are important in the theory of variational formulations of partial differential equations.

Let $\Omega \subset \mathbb{R}^d$ be a domain and $p \in (1,\infty)$ mit $p^{-1} + q^{-1} = 1$. □

Definition 3.30 The space $W^{-k,q}(\Omega)$. The space $W^{-k,q}(\Omega), k \in \mathbb{N} \cup \{0\}$, contains distributions which are defined on $W^{k,p}(\Omega)$

$$W^{-k,q}(\Omega) = \left\{ \varphi \in (C_0^\infty(\Omega))' \;:\; \|\varphi\|_{W^{-k,q}} < \infty \right\}$$

with

$$\|\varphi\|_{W^{-k,q}} = \sup_{u \in C_0^\infty(\Omega), u \neq 0} \frac{\langle \varphi, u \rangle}{\|u\|_{W^{k,p}(\Omega)}}.$$

□

Remark 3.31 *On the spaces $W^{-k,p}(\Omega)$.*

- It is $W^{-k,q}(\Omega) = \left[W_0^{k,p}(\Omega) \right]'$, i.e., $W^{-k,q}(\Omega)$ can be identified with the dual space of $W_0^{k,p}(\Omega)$. In particular it is $H^{-1}(\Omega) = \left(H_0^1(\Omega) \right)'$.
- It is

$$\ldots \subset W^{2,p}(\Omega) \subset W^{1,p}(\Omega) \subset L^p(\Omega) \subset W^{-1,q}(\Omega) \subset W^{-2,q}(\Omega) \ldots$$

□

Definition 3.32 Sobolev–Slobodeckij space. Let $s \in \mathbb{R}$, then the Sobolev–Slobodeckij or Sobolev space $H^s(\Omega)$ is defined as follows:

- $s \in \mathbb{Z}$. $H^s(\Omega) = W^{s,2}(\Omega)$.
- $s > 0$ with $s = k + \sigma$, $k \in \mathbb{N} \cup \{0\}$, $\sigma \in (0,1)$. The space $H^s(\Omega)$ contains all functions u for which the following norm is finite:

$$\|u\|_{H^s(\Omega)}^2 = \|u\|_{H^k(\Omega)}^2 + |u|_{k+\sigma}^2 \,,$$

with

$$\begin{aligned} (u,v)_{H^s(\Omega)} &= (u,v)_{H^k} + (u,v)_{k+\sigma}, \quad |u|_{k+\sigma}^2 = (u,u)_{k+\sigma}, \\ (u,v)_{k+\sigma} &= \sum_{|\boldsymbol{\alpha}|=k} \int_\Omega \int_\Omega \frac{\left(D^{\boldsymbol{\alpha}} u(\mathbf{x}) - D^{\boldsymbol{\alpha}} u(\mathbf{y})\right)\left(D^{\boldsymbol{\alpha}} v(\mathbf{x}) - D^{\boldsymbol{\alpha}} v(\mathbf{y})\right)}{\|\mathbf{x}-\mathbf{y}\|_2^{d+2\sigma}} \, d\mathbf{x} d\mathbf{y}, \end{aligned}$$

- $s < 0$. $H^s(\Omega) = \left[H_0^{-s}(\Omega) \right]'$ with $H_0^{-s}(\Omega) = \overline{C_0^\infty(\Omega)}^{\|\cdot\|_{H^{-s}(\Omega)}}$.

□

3.6 Theorem on Equivalent Norms

Definition 3.33 Equivalent norms. Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on the linear space X are said to be equivalent if there are constants C_1 and C_2 such that

$$C_1 \|u\|_1 \leq \|u\|_2 \leq C_2 \|u\|_1 \ \forall\, u \in X.$$

□

Remark 3.34 *On equivalent norms.* Many important properties, like continuity or convergence, do not change if an equivalent norm is considered. □

Theorem 3.35 Equivalent norms in $W^{k,p}(\Omega)$. *Let $\Omega \subset \mathbb{R}^d$ be a domain with Lipschitz boundary Γ, $p \in [1,\infty]$, and $k \in \mathbb{N}$. Let $\{f_i\}_{i=1}^l$ be a system with the following properties:*
1) $f_i : W^{k,p}(\Omega) \to \mathbb{R}_+ \cup \{0\}$ is a semi norm,
2) $\exists C_i > 0$ with $0 \leq f_i(v) \leq C_i \|v\|_{W^{k,p}(\Omega)}$, $\forall\, v \in W^{k,p}(\Omega)$,
3) f_i is a norm on the polynomials of degree $k-1$, i.e., if for $v \in P_{k-1} = \left\{\sum_{|\boldsymbol{\alpha}|\leq k-1} C_{\boldsymbol{\alpha}} x^{\boldsymbol{\alpha}}\right\}$ it holds that $f_i(v) = 0$, $i = 1,\ldots,l$, then it is $v \equiv 0$.
Then, the norm $\|\cdot\|_{W^{k,p}(\Omega)}$ defined in (3.6) and the norm

$$\begin{aligned} \|u\|'_{W^{k,p}(\Omega)} &:= \left(\sum_{i=1}^{l} f_i^p(u) + |u|^p_{W^{k,p}(\Omega)}\right)^{1/p} \quad \text{with} \\ |u|_{W^{k,p}(\Omega)} &= \left(\sum_{|\boldsymbol{\alpha}|=k} \int_\Omega |D^{\boldsymbol{\alpha}} u(\mathbf{x})|^p \ d\mathbf{x}\right)^{1/p} \end{aligned}$$

are equivalent.

Remark 3.36 *On semi norms.* For a semi norm $f_i(\cdot)$, one cannot conclude from $f_i(v) = 0$ that $v = 0$. The third assumptions however states, that this conclusion can be drawn for all polynomials up to a certain degree. □

Example 3.37 *Equivalent norms in Sobolev spaces.*

- The following norms are equivalent to the standard norm in $W^{1,p}(\Omega)$:

$$\begin{aligned} \text{a)} \quad \|u\|'_{W^{1,p}(\Omega)} &= \left(\left|\int_\Omega u \ d\mathbf{x}\right|^p + |u|^p_{W^{1,p}(\Omega)}\right)^{1/p}, \\ \text{b)} \quad \|u\|'_{W^{1,p}(\Omega)} &= \left(\left|\int_\Gamma u \ d\mathbf{s}\right|^p + |u|^p_{W^{1,p}(\Omega)}\right)^{1/p}, \\ \text{c)} \quad \|u\|'_{W^{1,p}(\Omega)} &= \left(\int_\Gamma |u|^p \ d\mathbf{s} + |u|^p_{W^{1,p}(\Omega)}\right)^{1/p}. \end{aligned}$$

- In $W^{k,p}(\Omega)$ it is

$$\|u\|'_{W^{k,p}(\Omega)} = \left(\sum_{i=0}^{k-1} \int_\Gamma \left|\frac{\partial^i u}{\partial \mathbf{n}^i}\right|^p \ d\mathbf{s} + |u|^p_{W^{k,p}(\Omega)}\right)^{1/p}$$

  equivalent to the standard norm. Here, $\mathbf{n}$ denotes the outer normal on Γ with $\|\mathbf{n}\|_2 = 1$.

- In the case $W_0^{k,p}(\Omega)$, one does not need the regularity of the boundary. It is

$$\|u\|'_{W_0^{k,p}(\Omega)} = |u|_{W^{k,p}(\Omega)},$$

i.e., in the spaces $W_0^{k,p}(\Omega)$ the standard semi norm is equivalent to the standard norm.
In particular, it is for $u \in H_0^1(\Omega)$ $(k = 1, p = 2)$

$$C_1 \|u\|_{H^1(\Omega)} \leq \|\nabla u\|_{L^2(\Omega)} \leq C_2 \|u\|_{H^1(\Omega)}.$$

It follows that there is a constant $C > 0$ such that

$$\|u\|_{L^2(\Omega)} \leq C \|\nabla u\|_{L^2(\Omega)} \quad \forall\, u \in H_0^1(\Omega). \tag{3.8}$$

□

3.7 Some Inequalities in Sobolev Spaces

Remark 3.38 *Motivation.* This section presents a generalization of the last part of Example 3.37. It will be shown that for inequalities of type (3.8) it is not necessary that the trace vanishes on the complete boundary.

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with boundary Γ and let $\Gamma_1 \subset \Gamma$ with $\text{meas}_{\mathbb{R}^{d-1}}(\Gamma_1) = \int_{\Gamma_1} d\mathbf{s} > 0$.

One considers the space

$$\begin{aligned} V_0 &= \left\{v \in W^{1,p}(\Omega) \,:\, v|_{\Gamma_1} = 0\right\} \subset W^{1,p}(\Omega) \text{ if } \Gamma_1 \subset \Gamma, \\ V_0 &= W_0^{1,p}(\Omega) \text{ if } \Gamma_1 = \Gamma \end{aligned}$$

with $p \in [1, \infty)$. □

Lemma 3.39 Friedrichs[5] inequality, Poincaré[6] inequality, Poincaré–Friedrichs inequality. *Let $p \in [1, \infty)$ and $\text{meas}_{\mathbb{R}^{d-1}}(\Gamma_1) > 0$. Then it is for all $u \in V_0$*

$$\int_\Omega |u(\mathbf{x})|^p \, d\mathbf{x} \leq C_P \int_\Omega \|\nabla u(\mathbf{x})\|_2^p \, d\mathbf{x},$$

where $\|\cdot\|_2$ is the Euclidean vector norm.

Proof: The inequality will be proved with the theorem on equivalent norms, Theorem 3.35. Let $f_1(u) \,:\, W^{1,p}(\Omega) \to \mathbb{R}_+ \cup \{0\}$ with

$$f_1(u) = \left(\int_{\Gamma_1} |u(\mathbf{s})|^p \, d\mathbf{s}\right)^{1/p}.$$

This functions has the following properties:
1) $f_1(u)$ is a semi norm.
2) It is

$$\begin{aligned} 0 &\leq f_1(u) = \left(\int_{\Gamma_1} |u(\mathbf{s})|^p \, d\mathbf{s}\right)^{1/p} \leq \left(\int_{\Gamma} |u(\mathbf{s})|^p \, d\mathbf{s}\right)^{1/p} \\ &= \|u\|_{L^p(\Gamma)} = \|\gamma u\|_{L^p(\Gamma)} \leq C \|u\|_{W^{1,p}(\Omega)}. \end{aligned}$$

The last estimate follows from the continuity of the trace operator.

[5] Friedrichs
[6] Poincaré

3) Let $v \in P_0$, i.e., v is a constant. Then, one obtains from

$$0 = f_1(v) = \left(\int_{\Gamma_1} |v(\mathbf{s})|^p \ d\mathbf{s} \right)^{1/p} = |v| \left(\text{meas}_{\mathbb{R}^{d-1}} (\Gamma_1) \right)^{1/p},$$

that $|v| = 0$.

Hence, all assumptions of Theorem 3.35 are satisfied. That means, there are two constants C_1 and $C_2$with

$$C_1 \underbrace{\left(\int_{\Gamma_1} |u(\mathbf{s})|^p \ d\mathbf{s} + \int_\Omega \|\nabla u(\mathbf{x})\|_2^p \ d\mathbf{x} \right)^{1/p}}_{\|u\|'_{W^{1,p}(\Omega)}} \leq \|u\|_{W^{1,p}(\Omega)} \leq C_2 \|u\|'_{W^{1,p}(\Omega)} \ \forall \ u \in W^{1,p}(\Omega).$$

In particular, it follows that

$$\int_\Omega |u(\mathbf{x})|^p \ d\mathbf{x} + \int_\Omega \|\nabla u(\mathbf{x})\|_2^p \ d\mathbf{x} \leq C_2^p \left(\int_{\Gamma_1} |u(\mathbf{s})|^p \ d\mathbf{s} + \int_\Omega \|\nabla u(\mathbf{x})\|_2^p \ d\mathbf{x} \right)$$

or

$$\int_\Omega |u(\mathbf{x})|^p \ d\mathbf{x} \leq C_P \left(\int_{\Gamma_1} |u(\mathbf{s})|^p \ d\mathbf{s} + \int_\Omega \|\nabla u(\mathbf{x})\|_2^p \ d\mathbf{x} \right)$$

with $C_P = C_2^p$. Since $u \in V_0$ vanishes on Γ_1, the statement of the lemma is proved. ■

Remark 3.40 *On the Poincaré–Friedrichs inequality.* In the space V_0 becomes $|\cdot|_{W^{1,p}}$ a norm which is equivalent to $\|\cdot\|_{W^{1,p}(\Omega)}$. The classical Poincaré–Friedrichs inequality is given for $\Gamma_1 = \Gamma$ and $p = 2$

$$\|u\|_{L^2} \leq C_P \|\nabla u\|_{L^2} \ \forall \ u \in H_0^1(\Omega),$$

where the constant depends only on the diameter of the domain Ω. □

Lemma 3.41 Another inequality of Poincaré–Friedrichs type. *Let $\Omega' \subset \Omega$ with $\text{meas}_{\mathbb{R}^d}(\Omega') = \int_{\Omega'} d\mathbf{x} > 0$, then for all $u \in W^{1,p}(\Omega)$ it is*

$$\int_\Omega |u(\mathbf{x})|^p \ d\mathbf{x} \leq C \left(\left| \int_{\Omega'} u(\mathbf{x}) \ d\mathbf{x} \right|^p + \int_\Omega \|\nabla u(\mathbf{x})\|_2^p \ d\mathbf{x} \right).$$

Proof: *Exercise.* ■

3.8 The Gaussian Theorem

Remark 3.42 *Motivation.* The Gaussian theorem is the generalization of the integration by parts from calculus. This operation is very important for the theory of weak or variational solutions of partial differential equations. One has to study, under which conditions on the regularity of the domain and of the functions it is well defined. □

Theorem 3.43 Gaussian theorem. *Let $\Omega \subset \mathbb{R}^d, d \geq 2$, be a bounded domain with Lipschitz boundary Γ. Then, the following identity holds for all $u \in W^{1,1}(\Omega)$*

$$\int_\Omega \partial_i u(\mathbf{x}) \ d\mathbf{x} = \int_\Gamma u(\mathbf{s}) \mathbf{n}_i(\mathbf{s}) \ d\mathbf{s}, \tag{3.9}$$

where $\mathbf{n}$ is the unit outer normal vector on Γ.

Proof: sketch. First of all, one proves the statement for functions from $C^1(\overline{\Omega})$. This proof is somewhat longer and it is referred to the literature, e.g., Evans (2010).

The space $C^1(\overline{\Omega})$ is dense in $W^{1,1}(\Omega)$, see Remark 3.21. Hence, for all $u \in W^{1,1}(\Omega)$ there is a sequence $\{u_n\}_{n=1}^{\infty} \in C^1(\overline{\Omega})$ with

$$\lim_{n\to\infty} \|u - u_n\|_{W^{1,1}(\Omega)} = 0$$

and (3.9) holds for all functions $u_n(\mathbf{x})$. It will be shown that the limit of the left hand side converges to the left hand side of (3.9) and the limit of the right hand side converges to the right hand side of (3.9).

From the convergence in $\|\cdot\|_{W^{1,1}(\Omega)}$, one has in particular

$$\lim_{n\to\infty} \int_\Omega \partial_i u_n(\mathbf{x})\ d\mathbf{x} = \int_\Omega \partial_i u(\mathbf{x})\ d\mathbf{x}.$$

On the other hand, the continuity of the trace operator gives

$$\lim_{n\to\infty} \|u - u_n\|_{L^1(\Gamma)} \le C \lim_{n\to\infty} \|u - u_n\|_{W^{1,1}} = 0,$$

from what follows that

$$\lim_{n\to\infty} \int_\Gamma u_n(\mathbf{s})\ d\mathbf{s} = \int_\Gamma u(\mathbf{s})\ d\mathbf{s}.$$

Since for a Lipschitz boundary, the normal $\mathbf{n}$ is almost everywhere continuous, it is

$$\lim_{n\to\infty} \int_\Gamma u_n(\mathbf{s})\mathbf{n}_i(\mathbf{s})\ d\mathbf{s} = \int_\Gamma u(\mathbf{s})\mathbf{n}_i(\mathbf{s})\ d\mathbf{s}.$$

Thus, the limits lead to (3.9). ■

Corollary 3.44 Vector field. *Let the conditions of Theorem 3.43 on the domain Ω be satisfied and let $\mathbf{u} \in \left(W^{1,1}(\Omega)\right)^d$ be a vector field. Then it is*

$$\int_\Omega \nabla \cdot \mathbf{u}(\mathbf{x})\ d\mathbf{x} = \int_\Gamma \mathbf{u}(\mathbf{s}) \cdot \mathbf{n}(\mathbf{s})\ d\mathbf{s}.$$

Proof: The statement follows by adding (3.9) from $i = 1$ to $i = d$. ■

Corollary 3.45 Integration by parts. *Let the conditions of Theorem 3.43 on the domain Ω be satisfied. Consider $u \in W^{1,p}(\Omega)$ and $v \in W^{1,q}(\Omega)$ with $p \in (1,\infty)$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then it is*

$$\int_\Omega \partial_i u(\mathbf{x}) v(\mathbf{x})\ d\mathbf{x} = \int_\Gamma u(\mathbf{s}) v(\mathbf{s}) \mathbf{n}_i(\mathbf{s})\ d\mathbf{s} - \int_\Omega u(\mathbf{x}) \partial_i v(\mathbf{x})\ d\mathbf{x}.$$

Proof: *exercise.* ■

Corollary 3.46 First Green[7]'s formula. *Let the conditions of Theorem 3.43 on the domain Ω be satisfied, then it is*

$$\int_\Omega \nabla u(\mathbf{x}) \cdot \nabla v(\mathbf{x})\ d\mathbf{x} = \int_\Gamma \frac{\partial u}{\partial \mathbf{n}}(\mathbf{s}) v(\mathbf{s})\ d\mathbf{s} - \int_\Omega \Delta u(\mathbf{x}) v(\mathbf{x})\ d\mathbf{x}$$

for all $u \in H^2(\Omega)$ and $v \in H^1(\Omega)$.

Proof: From the definition of the Sobolev spaces it follows that the integrals are well defined. Now, the proof follows the proof of Corollary 3.45, where one has now to sum over the components. ■

[7] Georg Green (1793 – 1841)

Remark 3.47 *On the first Green's formula.* The first Green's formula is the formula of integrating by parts once. The boundary integral can be equivalently written in the form

$$\int_\Gamma \nabla u(\mathbf{s}) \cdot \mathbf{n}(\mathbf{s}) v(\mathbf{s}) \, d\mathbf{s}.$$

The formula of integrating by parts twice is called second Green's formula. □

Corollary 3.48 Second Green's formula. *Let the conditions of Theorem 3.43 on the domain Ω be satisfied, then one has*

$$\int_\Omega \left(\Delta u(\mathbf{x}) v(\mathbf{x}) - \Delta v(\mathbf{x}) u(\mathbf{x}) \right) \, d\mathbf{x} = \int_\Gamma \left(\frac{\partial u}{\partial \mathbf{n}}(\mathbf{s}) v(\mathbf{s}) - \frac{\partial v}{\partial \mathbf{n}}(\mathbf{s}) u(\mathbf{s}) \right) \, d\mathbf{s}$$

for all $u, v \in H^2(\Omega)$.

3.9 Sobolev Imbedding Theorems

Remark 3.49 *Motivation.* This section studies the question which Sobolev spaces are subspaces of other Sobolev spaces. With this property, called imbedding, it is possible to estimate the norm of a function in the subspace by the norm in the larger space. □

Lemma 3.50 Imbedding of Sobolev spaces with same integration power p and different orders of the derivative. *Let $\Omega \subset \mathbb{R}^d$ be a domain with $p \in [1, \infty)$ and $k \leq m$, then it is $W^{m,p}(\Omega) \subset W^{k,p}(\Omega)$.*

Proof: The statement of this lemma follows directly from the definition of Sobolev spaces, see Definition 3.20. ■

Lemma 3.51 Imbedding of Sobolev spaces with the same order of the derivative k and different integration powers. *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain, $k \geq 0$, and $p, q \in [1, \infty]$ with $q > p$. Then it is $W^{k,q}(\Omega) \subset W^{k,p}(\Omega)$.*

Proof: *exercise.* ■

Remark 3.52 *Imbedding of Sobolev spaces with the same order of the derivative k and the same integration power p in imbedded domains.* Let $\Omega \subset \mathbb{R}^d$ be a domain with sufficiently smooth boundary Γ, $k \geq 0$, and $p \in [1, \infty]$. Then there is a map $E : W^{k,p}(\Omega) \to W^{k,p}(\mathbb{R}^d)$, the so-called (simple) extension, with

- $Ev|_\Omega = v$,
- $\|Ev\|_{W^{k,p}(\mathbb{R}^d)} \leq C \|v\|_{W^{k,p}(\Omega)}$, with $C > 0$,

e.g., see (Adams, 1975, Chapter IV) for details. Likewise, the natural restriction $e : W^{k,p}(\mathbb{R}^d) \to W^{k,p}(\Omega)$ can be defined and it is $\|ev\|_{W^{k,p}(\Omega)} \leq \|v\|_{W^{k,p}(\mathbb{R}^d)}$. □

Theorem 3.53 A Sobolev inequality. *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with Lipschitz boundary Γ, $k \geq 0$, and $p \in [1, \infty)$ with*

$$\begin{array}{ll} k \geq d & \text{for } p = 1, \\ k > d/p & \text{for } p > 1. \end{array}$$

Then there is a constant C such that for all $u \in W^{k,p}(\Omega)$ it follows that $u \in C_B(\Omega)$, where

$$C_B(\Omega) = \{ v \in C(\Omega) \ : \ v \text{ is bounded} \},$$

and it is

$$\|u\|_{C_B(\Omega)} = \|u\|_{L^\infty(\Omega)} \leq C \|u\|_{W^{k,p}(\Omega)}. \tag{3.10}$$

Proof: See literature, e.g., Adams (1975); Adams and Fournier (2003). ■

Remark 3.54 *On the Sobolev inequality.* The Sobolev inequality states that each function with sufficiently many weak derivatives (the number depends on the dimension of Ω and the integration power) can be considered as a continuous and bounded function in Ω. One says that $W^{k,p}(\Omega)$ is imbedded in $C_B(\Omega)$. It is

$$C\left(\overline{\Omega}\right) \subset C_B(\Omega) \subset C(\Omega).$$

Consider $\Omega = (0,1)$ and $f_1(x) = 1/x$ and $f_2(x) = \sin(1/x)$. Then, $f_1 \in C(\Omega)$, $f_1 \notin C_B(\Omega)$ and $f_2 \in C_B(\Omega)$, $f_2 \notin C(\overline{\Omega})$.

Of course, it is possible to apply this theorem to weak derivatives of functions. Then, one obtains imbeddings like $W^{k,p}(\Omega) \to C_B^s(\Omega)$ for $(k-s)p > d, p > 1$. A comprehensive overview on imbeddings can be found in Adams (1975); Adams and Fournier (2003). □

Example 3.55 $H^1(\Omega)$ *in one dimension.* Let $d = 1$ and Ω be a bounded interval. Then, each function from $H^1(\Omega)$ $(k = 1, p = 2)$ is continuous and bounded in Ω. □

Example 3.56 $H^1(\Omega)$ *in higher dimensions.* The functions from $H^1(\Omega)$ are in general not continuous for $d \geq 2$. This property will be shown with the following example.

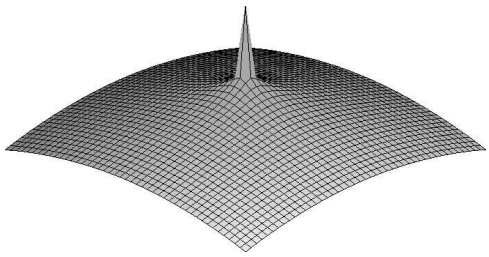


Figure 3.4: The function $f(\mathbf{x})$ of Example 3.56 for $d = 2$.

Let $\Omega = \{\mathbf{x} \in \mathbb{R}^d \ : \ \|\mathbf{x}\|_2 < 1/2\}$ and $f(\mathbf{x}) = \ln|\ln\|\mathbf{x}\|_2|$, see Figure 3.4. For $\|\mathbf{x}\|_2 < 1/2$ it is $|\ln\|\mathbf{x}\|_2| = -\ln\|\mathbf{x}\|_2$ and one gets for $\mathbf{x} \neq \mathbf{0}$

$$\partial_i f(\mathbf{x}) = -\frac{1}{\ln\|\mathbf{x}\|_2}\frac{1}{\|\mathbf{x}\|_2}\frac{x_i}{\|\mathbf{x}\|_2} = -\frac{x_i}{\|\mathbf{x}\|_2^2 \ln\|\mathbf{x}\|_2}.$$

For $p \leq d$, one obtains

$$\left|\frac{\partial f}{\partial x_i}(\mathbf{x})\right|^p = \underbrace{\left|\frac{x_i}{\|\mathbf{x}\|_2}\right|^p}_{\leq 1}\underbrace{\left|\frac{1}{\|\mathbf{x}\|_2 \ln\|\mathbf{x}\|_2}\right|^p}_{\geq e} \leq \left|\frac{1}{\|\mathbf{x}\|_2 \ln\|\mathbf{x}\|_2}\right|^d.$$

The estimate of the second factor can be obtained, e.g., with a discussion of the curve. Using now spherical coordinates, $\rho = e^{-t}$ and S^{d-1} is the unit sphere, yields

$$\begin{aligned}\int_\Omega |\partial_i f(\mathbf{x})|^p \, d\mathbf{x} &\leq \int_\Omega \frac{d\mathbf{x}}{\|\mathbf{x}\|_2^d \left|\ln\|\mathbf{x}\|_2\right|^d} = \int_{S^{d-1}}\int_0^{1/2} \frac{\rho^{d-1}}{\rho^d \left|\ln\rho\right|^d} \, d\rho d\omega \\ &= \text{meas}\left(S^{d-1}\right)\int_0^{1/2} \frac{d\rho}{\rho\left|\ln\rho\right|^d} = -\text{meas}\left(S^{d-1}\right)\int_\infty^{\ln 2} \frac{dt}{t^d} < \infty,\end{aligned}$$

because of $d \geq 2$.

It follows that $\partial_i f \in L^p(\Omega)$ with $p \leq d$. Analogously, one proves that $f \in L^p(\Omega)$ with $p \leq d$. Altogether, one has $f \in W^{1,p}(\Omega)$ with $p \leq d$. However, it is $f \notin L^\infty(\Omega)$. This example shows that the condition $k > d/p$ for $p > 1$ is sharp.

In particular, it was proved for $p = 2$ that from $f \in H^1(\Omega)$ in general it does not follow that $f \in C(\Omega)$. □

Example 3.57 *The assumption of a Lipschitz boundary.* Also the assumption that Ω is a Lipschitz domain is of importance.

Consider $\Omega = \{(x, y) \in \mathbb{R}^2 \ : \ 0 < x < 1, \ |y| < x^r, r > 1\}$, see Figure 3.5 for $r = 2$.

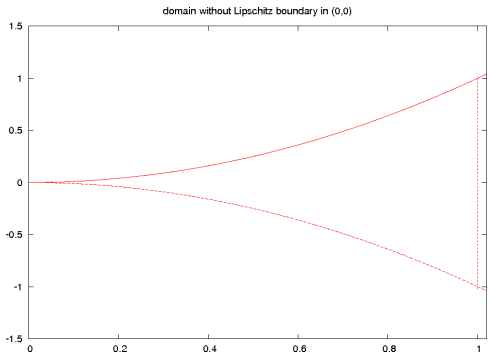


Figure 3.5: Domain of Example 3.57.

For $u(x, y) = x^{-\varepsilon/p}$ with $0 < \varepsilon < r$ it is

$$\partial_x u = x^{-\varepsilon/p-1} \left(-\frac{\varepsilon}{p}\right) = C(\varepsilon, p) x^{-\varepsilon/p-1}, \ \partial_y u = 0.$$

It follows that

$$\begin{aligned} \sum_{|\boldsymbol{\alpha}|=1} \int_\Omega |D^{\boldsymbol{\alpha}} u(x, y)|^p \ dxdy &= C(\varepsilon, p) \int_\Omega x^{-\varepsilon-p} \ dxdy \\ &= C(\varepsilon, p) \int_0^1 x^{-\varepsilon-p} \left(\int_{-x^r}^{x^r} dy \right) dx \\ &= \tilde{C}(\varepsilon, p) \int_0^1 x^{-\varepsilon-p+r} \ dx. \end{aligned}$$

This value is finite for $-\varepsilon - p + r > -1$ or for $p < 1 + r - \varepsilon$, respectively. If one chooses $r \geq \varepsilon > 0$, then it is $u \in W^{1,p}(\Omega)$. But for $\varepsilon > 0$ the function $u(\mathbf{x})$ is not bounded in Ω, i.e., $u \notin L^\infty(\Omega)$.

The unbounded values of the function are compensated in the integration by the fact that the neighborhood of the singular point $(0, 0)$ possesses a small measure. □